The Islamia University of Bahawalpur

Department of Information Technology

SOFTWARE REQUIREMENTS SPECIFICATION
(SRS DOCUMENT)

for

Group Study Room with Chat

By

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ADP INFORMATION TECHNOLOGY

Revision History

Name	Date	Reason for Changes	Version

Application Evaluation History

Comments (by committee) – include the ones given at scope time both in doc and presentation	Action Taken

Supervised by: Sir Faisal Shahzad

Signature: ____________________________

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1. Introduction

The Group Study Room with Chat is a web-based application developed to enhance academic collaboration among students. The following points outline the context and background of the system:

- In today's hybrid learning environment, the need for virtual study spaces has grown significantly.
- Students often find it challenging to replicate the dynamics of in-person group study sessions.
- This platform bridges that gap by offering a user-friendly, accessible environment where students can engage in real-time academic discussions, collaborate, and share study materials.
- The application facilitates peer-to-peer learning by allowing students to create or join topic-specific study rooms, communicate through chat, and share various study materials such as PDFs, documents, and links.
- This project is developed by Hoorain Hasnain as part of the final year project for the BS in Information Technology at The Islamia University of Bahawalpur, under the guidance of Sir Faisal Shahzad.
- The system aims to serve the academic community of over 5,000 undergraduate students across various departments, offering them an easy-to-use platform for collaborative learning.

1.1. Purpose

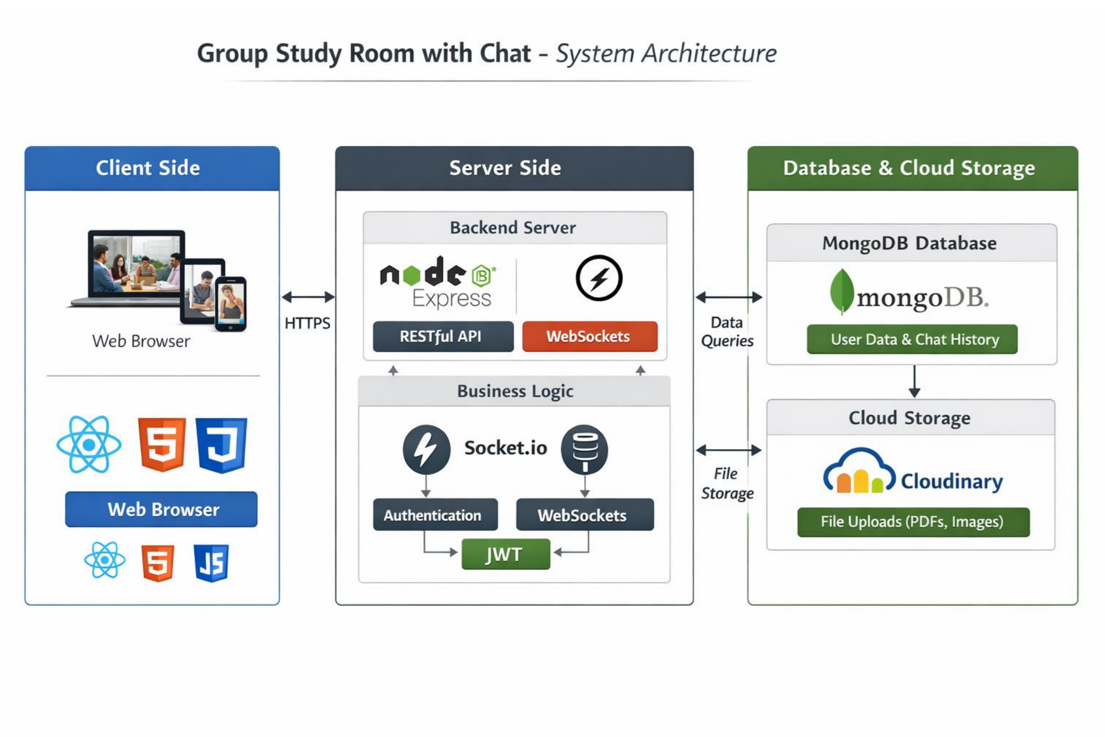
The purpose of this Software Requirements Specification (SRS) document is to provide a clear and concise outline of all functional and non-functional requirements for the Group Study Room with Chat system. Key objectives include:

- This document is aimed at developers, system administrators, and academic supervisors involved in the development, implementation, and maintenance of the platform.
- The SRS serves as a guide for understanding the goals, expectations, and design considerations for the system.
- The primary objective is to create an online group study environment where students can efficiently collaborate, share resources, and engage in academic discussions in real-time.
- The platform will allow students to communicate through a live chat feature, share files, and access a variety of study resources in a minimalistic, easy-to-navigate interface.
- The platform will also feature simple user authentication to ensure that only registered users can access study rooms.

1.2. Scope

The Group Study Room with Chat system will be a fully web-based platform, targeting university students who wish to collaborate and engage in group study sessions. The scope includes:

- Users will have the ability to create study rooms, join existing rooms, chat in real time, and share study materials.
- Key features include user registration, authentication, room creation, and real-time chat with file sharing capabilities.
- The system will initially support text-based communication and file sharing within the study rooms.
- Future expansions may include additional features such as voice chat, video conferencing, and virtual whiteboards.
- The system will be designed for use on desktop and tablet browsers, with mobile responsiveness included.
- Mobile applications (iOS/Android) are deferred for future versions.



1.3. Product Perspective

The Group Study Room with Chat is designed as a standalone web application with the following architectural characteristics:

- The system does not depend on any existing university Learning Management System (LMS).
- It will be deployed on a cloud platform, ensuring ease of access and scalability for a large number of users.
- The system follows a modular architecture, with a front-end React-based interface, a backend powered by Node.js and Express, and a MongoDB database for data storage.

- The system’s goal is to complement existing academic tools by offering a simple, distraction-free environment tailored for student collaboration.
- It focuses on ease of use, low system requirements, and is optimized for group study dynamics rather than being an all-encompassing academic platform like LMS systems.

1.4. User Characteristics

The Group Study Room with Chat system will have two main types of users:

- **Students (Primary Users):** The system is primarily designed for undergraduate and postgraduate students enrolled at The Islamia University of Bahawalpur. These users will have basic digital literacy, meaning they can navigate web browsers, upload files, and participate in real-time text chats. No advanced technical knowledge is assumed, and the system is designed to be intuitive and easy to use.
- **Admin (Secondary Users):** Administrators will have more advanced digital literacy and will be responsible for managing the study rooms, moderating content, and monitoring system activity. Admins will access a dedicated dashboard to manage the platform’s features, such as creating or deleting rooms, generating reports, and overseeing user activities.

1.5. Similar Apps and Systems / Literature Review

Several platforms already exist that offer collaboration features such as chat and file sharing. A comparison is outlined below:

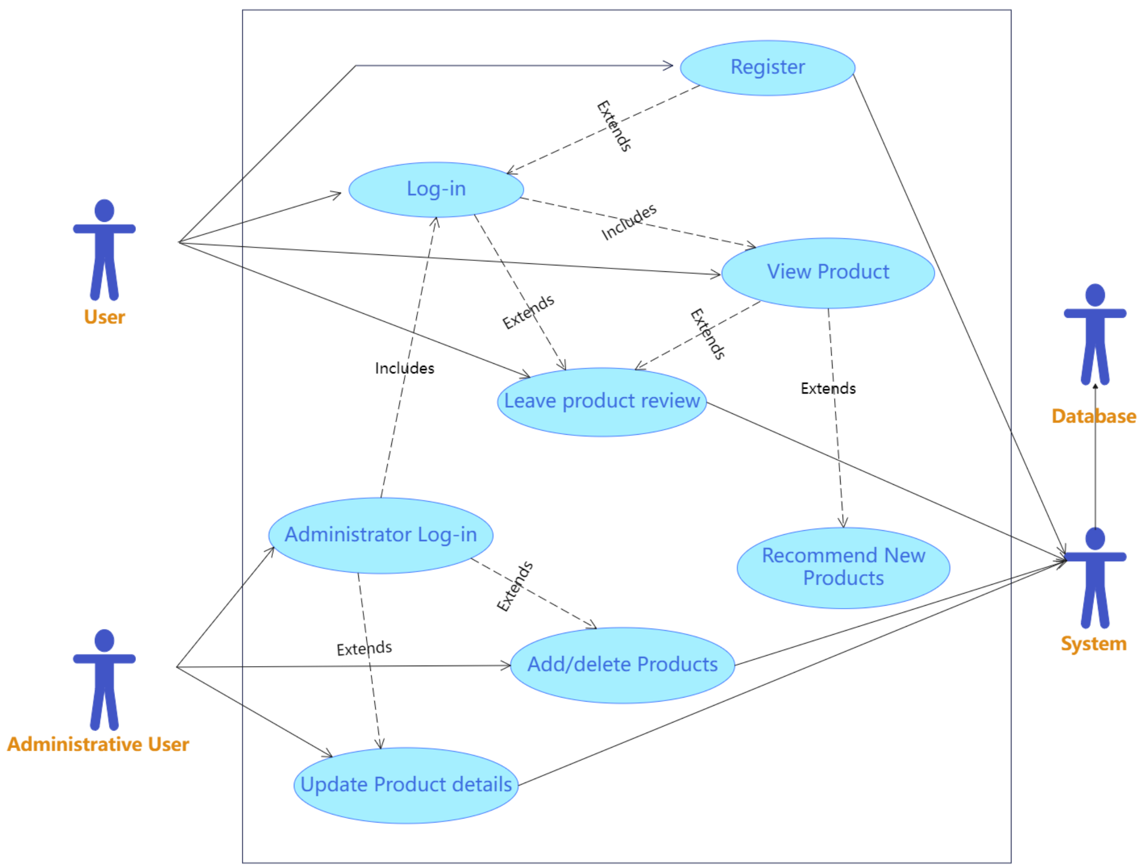
- **Discord:** Offers a feature-rich experience with voice, video, and text communication, but is cluttered and not tailored to academic workflows.
- **Microsoft Teams:** Provides collaboration tools but requires a Microsoft account and a more complex setup.
- **Google Meet:** Focuses primarily on video conferencing, with no persistent rooms or file libraries.
- In contrast, the Group Study Room with Chat offers a minimal, distraction-free environment specifically tailored for academic use.
- The system fills a niche by combining real-time text communication with persistent file sharing, making it a better solution for students seeking an organized, focused study environment.

1.6. Proposed Technologies

The Group Study Room with Chat system will be developed using a modern technology stack, chosen for its suitability for real-time applications and web-based development:

- **Frontend:** The user interface will be developed using HTML5, CSS3, and JavaScript. The system will leverage React.js for building a responsive, single-page web application that provides a smooth user experience.

- **Backend:** The backend will be powered by Node.js (v20 LTS) with Express.js as the RESTful API framework. Socket.io will be used to enable real-time communication within the study rooms.
- **Database:** The system will use MongoDB as the NoSQL database to store user data, study room information, and chat history.
- **Authentication:** JSON Web Tokens (JWT) will be used for user authentication, providing a secure method for handling user logins and maintaining session states.
- **File Storage:** Cloudinary will be used to store media files, and GridFS will handle larger document storage requirements.
- **Deployment:** The application will be hosted on Render.com for the backend and Vercel for the frontend, with GitHub Actions integrated for continuous deployment.



2. Requirements

The Group Study Room with Chat system is designed to address the needs of students who wish to engage in collaborative, virtual study sessions. The following points summarize this section:

- The functional requirements describe the core features of the system.
- The non-functional requirements define the quality attributes and operational constraints.

2.1. Functional Requirements

Functional requirements define the primary features and behavior of the system that directly align with the users' needs. Each requirement specifies a feature that must be implemented to fulfill the objectives of the platform.

2.1.1. Sign Up

Name: FR001

Purpose: The sign-up process allows new users to create an authenticated account on the platform, enabling them to access all study room features and become part of the virtual study community.

User(s): Student, Admin

Input:

- Full Name: The legal name of the user (minimum 3 characters).
- Email Address: A valid email format (e.g., user@domain.com).
- Password: Must be at least 8 characters long and include at least one uppercase letter, one digit, and one special character.
- Confirm Password: Must match the password exactly.
- Role Selection: User can choose between "Student" or "Admin" (Admin requires supervisor approval).

Output:

- A new user account is created.
- A JWT token is issued for authentication.
- The user is redirected to the main dashboard after successful registration.

2.2. Non-Functional Requirements

Non-functional requirements define the quality attributes, operational constraints, and system behaviors that ensure the system meets performance, reliability, security, and usability standards.

NFR01 – Performance

- The page load time must not exceed 3 seconds on a 10 Mbps connection.
- Chat messages must be delivered in less than 500 ms under normal network conditions.

NFR02 – Scalability

- The system must be capable of supporting a minimum of 500 concurrent WebSocket connections without degradation in performance.
- The architecture must allow for horizontal scaling through container orchestration, ensuring future growth without performance loss.

NFR03 – Security

- All passwords must be hashed using bcrypt (salt rounds ≥ 12).
- JWT validation must protect all API endpoints.
- All connections must be secured using HTTPS (TLS 1.3) to ensure data security and privacy.

NFR04 – Reliability

- The system must maintain an uptime of ≥ 99.5% per month to ensure consistent availability.
- Automated MongoDB backups must run daily with a 7-day retention policy to safeguard data.

NFR05 – Usability

- A new user should be able to complete registration and join their first study room within 3 minutes without needing external documentation.
- The system must comply with WCAG 2.1 Level AA accessibility standards to ensure it is usable by people with disabilities.

NFR06 – Maintainability

- The system's codebase must follow the MVC architecture for modularity and ease of maintenance.
- Inline JSDoc comments must be included in all modules to facilitate easier understanding and updates.
- Code coverage must be ≥ 70% through unit tests (using Jest) to ensure system reliability and maintainability.

NFR07 – Portability

- The application must render correctly on the latest versions of Chrome (120+), Firefox (120+), Edge (120+), and Safari (17+) on both desktop and tablet devices.

NFR08 – Data Integrity

- All database writes must be done using MongoDB transactions when atomicity is required to maintain consistent data.

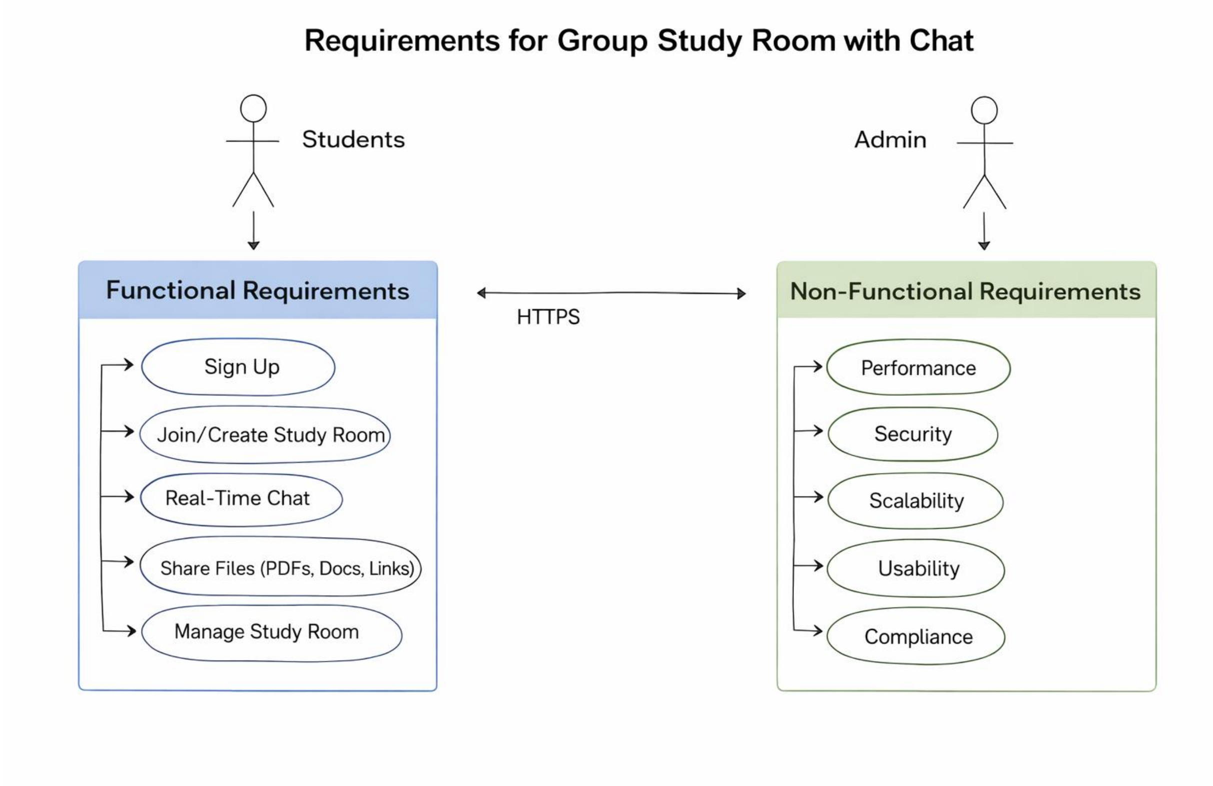
- Orphaned room references must be automatically cleaned up via scheduled jobs.

NFR09 – Compliance

- The system must comply with data privacy guidelines, including obtaining user consent for data storage during registration.
- Account deletion must fully remove all personally identifiable information (PII) from the system.

NFR10 – Localization

- Version 1.0 will be English-only; however, the codebase should use i18n-compatible string management to facilitate future localization into languages like Urdu.



Functional Requirements Specification

ID	Name	Purpose	User(s)	Input	Output
FR 001	Sign Up	Allows new users to create authenticated accounts	Student, Admin	Full Name, Email, Password, Confirm Password, Role Selection	Account created, JWT issued, redirect to dashboard
FR 002	Login	Authenticates existing users	Student, Admin	Email, Password	JWT validated, access granted
FR 003	Create Study Room	Enables users to create new rooms	Student	Room Name, Topic	Room created in database
FR 004	Join Study Room	Allows users to join existing rooms	Student	Room Code or Name	User added to room
FR 005	Chat in Real Time	Facilitates live communication	Student, Admin	Message input	Message displayed instantly via Socket.io
FR 006	Upload/Share Files	Enables sharing of study materials	Student	File (PDF, Doc, Link)	File stored in Cloudinary/GridFS
FR 007	Manage Rooms	Allows admins to moderate rooms	Admin	Room ID, Action (Delete/Edit)	Room updated or removed

Non-Functional Requirements Specification

ID	Category	Description	Metric / Standard
NFR01	Performance	Page load ≤ 3s; Chat delivery ≤ 500ms	10 Mbps connection
NFR02	Scalability	Support ≥ 500 WebSocket connections	Horizontal scaling via containers
NFR03	Security	bcrypt hashing (≥ 12 salt rounds); JWT validation; HTTPS TLS 1.3	OWASP standards
NFR04	Reliability	Uptime ≥ 99.5%; daily MongoDB backups (7-day retention)	SLA monitoring
NFR05	Usability	Registration ≤ 3 min; WCAG 2.1 AA compliance	Accessibility testing
NFR06	Maintainability	MVC architecture; JSDoc comments; ≥ 70% unit test coverage	Jest reports
NFR07	Portability	Works on latest Chrome, Firefox, Edge, Safari	Browser testing
NFR08	Data Integrity	MongoDB transactions for atomic writes; cleanup jobs	Scheduled scripts

ID	Category	Description	Metric / Standard
NFR09	Compliance	GDPR-style data consent; PII deletion on account removal	Privacy audit
NFR10	Localization	English v1.0; i18n ready for Urdu and others	Translation framework

3. Use Cases and Flow of Processes

This section outlines the formal Use Case descriptions for the Group Study Room with Chat system. The following points summarize the purpose of this section:

- Use cases define the system's interactions from the perspective of the user.
- They detail the processes and functionalities the system must support.
- Each use case provides a clear flow of events to achieve the desired outcomes.

3.1. Use Case 1: User Registration (Sign Up)

ID: UC001

Name: User Registration

Description: This use case describes the process by which a new user (student or admin) creates an authenticated account on the platform, allowing access to study rooms and all associated features.

Requirement(s): FR001

Actor(s): Student, Admin

Precondition:

- The user must have a valid email address and not be already registered on the platform.
- The user must not have any existing account in the system.

Postcondition:

- A new user account is created, stored in the database, and authenticated.
- The user receives a welcome notification.
- The user is redirected to the main dashboard.

Basic Flow:

- The user navigates to the Sign-Up page.
- The user enters their Full Name, Email Address, Password, and Confirm Password.
- The system validates all input fields: ensures Full Name has at least 3 characters, Email Address follows a valid email format, and Password is at least 8 characters with one uppercase letter, one digit, and one special character.
- The system checks that the Confirm Password field matches the Password field exactly.
- The system validates that the email address is unique and not already registered.
- The system hashes the password using bcrypt for secure storage.
- The system creates a new user document in MongoDB.
- The system generates a JWT token for the user to authenticate them for future sessions.
- The user is redirected to the main dashboard.

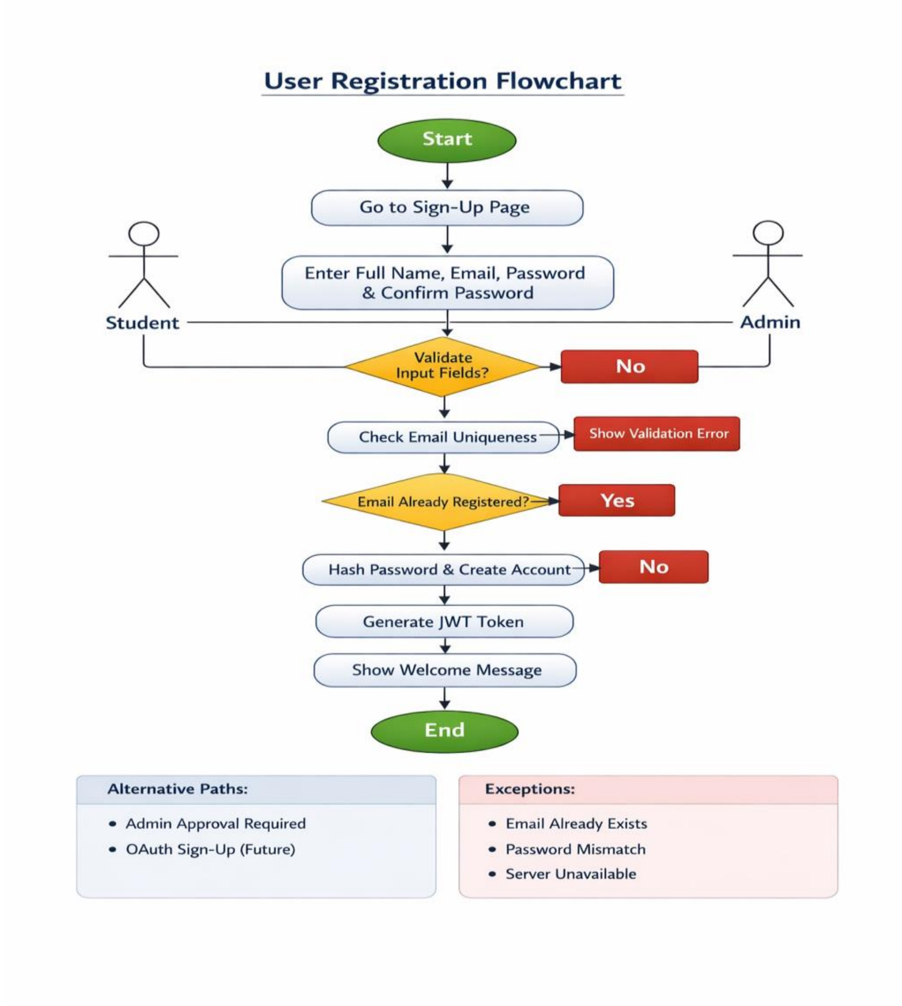
Alternative Flow:

- **Admin Sign-Up:** If the user is registering as an admin, they will be required to get approval from a supervisor.

- **OAuth Sign-Up:** A future feature could allow users to sign up via Google OAuth for faster registration.

Exceptions:

- **Email Already Registered:** If the email is already associated with an existing account, the system displays an error message: “An account with this email already exists. Please log in.”
- **Password Mismatch:** If the passwords do not match, the system highlights the Confirm Password field with an error message.
- **Server Unavailable:** If the server is down or unavailable, the system displays an error message: “Service temporarily unavailable. Please try again later.”



4. References

The following references provide foundational resources for the development of the Group Study Room with Chat system:

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